

Traceability Matrix

Restaurant Management System

Team 10 - Software Testing Project 2026

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Project	Restaurant Management System

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1 Introduction

This Traceability Matrix maps each requirement from the SRS to the corresponding test cases. It ensures:

- All requirements have at least one test case
- Test coverage can be measured
- Missing or untested requirements are identified
- Changes to requirements can be traced to affected tests

1.1 Status Legend

Status	Meaning
Pass	Test case executed and passed
Fail	Test case executed and failed (incident report created)
Partial	Mixed results - some test cases passed, some failed
N/A	Requirement not testable (deferred to assumptions)

2 User Authentication Requirements (SRS Section 7.1)

Req ID	Requirement Description	Test Case ID(s)	Status	Incident ID
REQ-7.1.1	Customer registration with email, password, name, phone	TC-RES-001, TC-RES-002	Pass	-
REQ-7.1.2	Authenticate users and create sessions upon login	TC-RES-003, TC-RES-004	Pass	-
REQ-7.1.3	Lock accounts for 15 minutes after three failed login attempts	TC-RES-005	Fail	INC-RES-001

3 Table Reservation Requirements (SRS Section 7.2)

Req ID	Requirement Description	Test Case ID(s)	Status	Incident ID
REQ-7.2.1	Display available time slots based on date, time, party size	TC-RES-006 to TC-RES-009	Partial	INC-RES-002, INC-RES-003
REQ-7.2.2	Automatically assign tables based on party size and capacity	TC-RES-011 to TC-RES-013	Fail	INC-RES-004
REQ-7.2.3	Allow pre-order selection during reservation	TC-RES-015	Pass	-
REQ-7.2.4	Create reservations with Pending Approval status	TC-RES-014, TC-RES-016, TC-RES-017	Fail	INC-RES-005, INC-RES-006
REQ-7.2.5	Allow customers to cancel reservations at least 24 hours in advance	TC-RES-018, TC-RES-019	Fail	INC-RES-007

4 Online Ordering Requirements (SRS Section 7.3)

Req ID	Requirement Description	Test Case ID(s)	Status	Incident ID
REQ-7.3.1	Display menu items by categories with details and pricing	TC-ORD-001, TC-ORD-002	Partial	INC-ORD-001
REQ-7.3.2	Maintain shopping cart state during user sessions	TC-ORD-003 to TC-ORD-007	Pass	-
REQ-7.3.3	Validate delivery addresses against service zones	TC-ORD-008 to TC-ORD-011	Partial	INC-ORD-002, INC-ORD-003
REQ-7.3.4	Process orders and generate unique order numbers	TC-ORD-014, TC-ORD-015	Partial	INC-ORD-004
REQ-7.3.5	Track order status through all stages	TC-ORD-016 to TC-ORD-020	Partial	INC-ORD-005 to INC-ORD-008

5 Event Booking Requirements (SRS Section 7.4)

Req ID	Requirement Description	Test Case ID(s)	Status	Incident ID
REQ-7.4.1	Check event space availability based on date, time, guest capacity	TC-EVT-002, TC-EVT-007	Partial	INC-EVT-002
REQ-7.4.2	Calculate catering costs based on guest count and menu items	TC-EVT-005	Pass	-
REQ-7.4.3	Prevent double-booking of event spaces	TC-EVT-003	Pass	-
REQ-7.4.4	Allow staff to approve/reject event requests with notifications	TC-EVT-009 to TC-EVT-011	Partial	INC-EVT-003

6 Administrative Functions (SRS Section 7.5)

Req ID	Requirement Description	Test Case ID(s)	Status	Incident ID
REQ-7.5.1	Allow admin to create, edit, delete menu items and categories	TC-ADM-004 to TC-ADM-007	Fail	INC-ADM-001
REQ-7.5.2	Allow table and event space configuration with capacity settings	TC-ADM-008, TC-ADM-009	Fail	INC-ADM-002
REQ-7.5.3	Allow admin to create staff accounts with role assignment	TC-ADM-010	Partial	INC-ADM-003

7 Non-Functional Requirements (SRS Section 8)

Req ID	Requirement Description	Test Case ID(s)	Status	Incident ID
REQ-8.1.1	Response time within 3 seconds	Manual timing	Pass	-
REQ-8.2.1	Zero double-booking incidents	TC-RES-010, TC-EVT-003	Pass	-
REQ-8.3.1	Authentication for all protected features	TC-LGN-003 to TC-LGN-005	Partial	INC-LGN-001
REQ-8.3.2	Role-based access control prevents unauthorized access	TC-LGN-006, TC-LGN-007	Fail	INC-LGN-002
REQ-8.3.3	Session timeout after 15 minutes inactivity	TC-RES-022	Fail	INC-RES-008
REQ-8.3.4	Passwords stored using secure hashing	Assumed (ASM-07)	N/A	-

8 Cross-Module & Workflow Requirements

Req ID	Requirement Description	Test Case ID(s)	Status	Incident ID
REQ-10.1	Customer makes reservation with pre-order workflow	TC-RES-006 to TC-RES-017	Partial	See Section 3
REQ-10.2	Customer places online order workflow	TC-ORD-001 to TC-ORD-025	Partial	See Section 4
REQ-10.3	Customer books event workflow	TC-EVT-001 to TC-EVT-012	Partial	See Section 5

9 Coverage Summary

Requirements Coverage

SRS Section	Requirements	Test Cases
Section 7.1 - User Authentication	3	5
Section 7.2 - Table Reservation	5	15
Section 7.3 - Online Ordering	5	25
Section 7.4 - Event Booking	4	7
Section 7.5 - Administrative Functions	3	6
Section 8 - Non-Functional	6	6
Section 10 - Workflows	3	All
Total	29	78

Coverage Percentage

Category	Coverage	Status
Functional Requirements	100%	All have test cases
Non-Functional Requirements	83%	Password hashing assumed
Workflow Requirements	100%	Covered by functional tests
Overall Coverage	95%	Target achieved

Test Execution Summary

Metric	Count	Percentage
Total Test Cases Executed	78	100%
Passed	47	60%
Failed	31	40%
Unique Bugs Found	24	-
Critical Bugs	1	4%
Major Bugs	13	54%
Minor Bugs	10	42%

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